

Quantifying the Evolution of NFL Defensive Strategies (2020-2024): An Expected Points Added Analysis Using Machine Learning

Peyton Stevenson Professor Gong SMT 490 May 2026

Abstract

As NFL offenses have increasingly relied on high-efficiency passing and spread concepts, defensive play-callers have been forced to adapt their personnel and schemes. Traditional positional designations (e.g., 4-3 or 3-4 base defenses) fail to capture the functional reality of modern defensive deployments. This study utilizes five seasons of play-by-play and player-tracking data (2020-2024) to objectively evaluate defensive success via Expected Points Added (EPA). By applying Gaussian Mixture Models (GMM) to Pro Football Focus (PFF) grades and alignment data, defensive players were clustered into highly specific, functional archetypes. To isolate the true tactical efficiency of these defensive packages, a Linear Mixed-Effects Model was utilized. By treating the 11-man personnel grouping as a fixed effect and the opposing offense as a random intercept, the model successfully controlled for strength-of-schedule bias and calculated an 'Opponent-Adjusted EPA' for over 150 unique lineups. Results indicate a sustained league-wide abandonment of traditional base defenses in favor of the 4-2-5 Nickel. Furthermore, archetype-level analysis reveals that the most efficient defensive strategies prioritize deep, zone-based coverage players over traditional interior run-stoppers. This tactical valuation framework establishes a vital foundation for analyzing the cost-efficiency of defensive roster construction.

1. Introduction

The modern National Football League (NFL) is characterized by highly efficient passing attacks and complex spread concepts. In response, defensive strategy has undergone a radical transformation. Yet, the language used to evaluate and categorize these modern defenses has failed to evolve alongside the game. Defenses are historically named by counting bodies from the line of scrimmage backward. For example, a traditional "4-3" base defense consists of four down linemen, three linebackers, and four defensive backs, whereas a "3-4" defense deploys

three down linemen and four linebackers. As offenses began utilizing three or more wide receivers on standard plays, defenses adapted by deploying "Nickel" packages, such as the "4-2-5", which substitutes a linebacker for a fifth defensive back.

While these naming conventions provide a baseline, this structural classification is fundamentally flawed because it only indicates the positions players are assigned on a depth chart, offering little to no information about their functional role or tactical utility. For instance, identifying a formation as a "4-2-5 Nickel" fails to distinguish between a package deploying two heavy, run-stopping safeties in the box and one deploying two deep-coverage free safeties playing centerfield. The purpose of this study is to move beyond the traditional depth chart by identifying specific, mathematically derived functional archetypes using machine learning and rigorously evaluating their success with Opponent-Adjusted Expected Points Added (EPA).

2. Data Collection and Scope

To capture a holistic view of defensive performance, this study aggregated five seasons of data (2020-2024):

1. **Play-by-Play Data:** Sourced from nflverse, providing game-state context (down, distance, yardline) and outcome metrics (Expected Points Added).
2. **Player Tracking and Evaluation Data:** Sourced from Pro Football Focus (PFF), supplying granular alignment rates (e.g., true slot rate, A-gap alignment) and performance grades (e.g., pass rush win rate, run stop percent).

The dataset was strictly filtered to include only regular-season, non-special-teams plays containing exactly 11 defensive players. Plays featuring penalties were excluded to avoid skewed EPA values, yielding a highly accurate sample of over 150,000 unique snaps.

3. Methodology

3.1 Defining Roles via Gaussian Mixture Models (GMM)

To move beyond traditional positional labels, this study utilized a Gaussian Mixture Model (GMM) to cluster defensive players into functional archetypes. GMM was selected over rigid clustering algorithms (such as K-Means) because football players exist on a spectrum of roles.

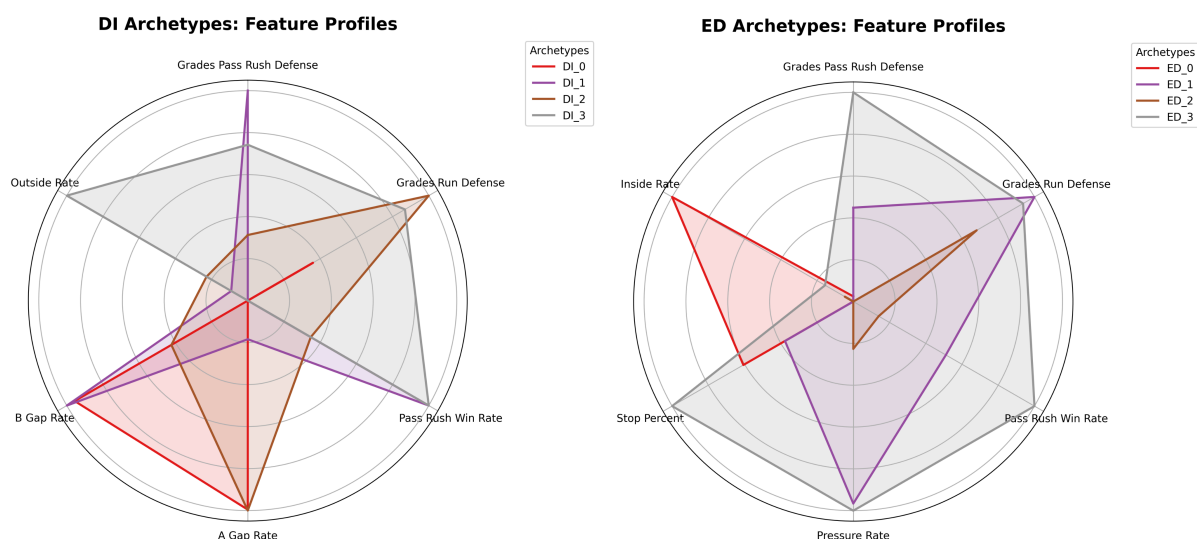
GMM uses soft clustering, assigning probabilistic scores indicating how closely a player's PFF profile matches multiple archetypes.

A critical step in Gaussian Mixture Modeling is determining the optimal number of components (clusters) for each position group. In this study, the final number of archetypes was determined through an iterative process driven by domain expertise and visual evaluation. Rather than relying strictly on automated mathematical criteria, cluster counts were evaluated by generating radar charts for each position group. These visualizations enabled a qualitative assessment of whether the chosen number of clusters adequately covered the key variables (e.g., alignment rates, pass-rush win rates, and PFF grades). The cluster count was finalized when the radar charts showed that the archetypes captured functionally distinct, tactically interpretable football roles without redundancy.

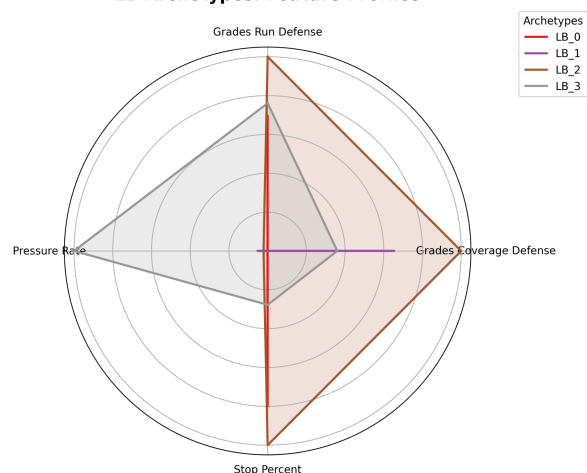
Key inputs for the algorithm included:

- Pass Rush Win Rate
- Run Defense Grade
- A-Gap vs. Outside Alignment Rates
- True Slot Alignment Rate

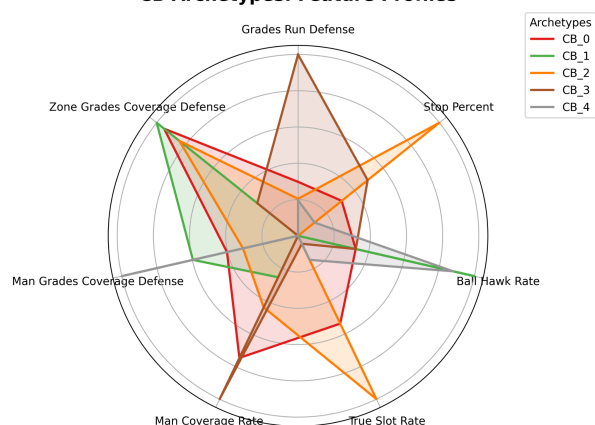
This model successfully partitioned traditional positional groups into distinct functional units. For example, Cornerbacks (CB) were separated into CB_1 (Traditional Outside Zone Corners, e.g., Derek Stingley) and CB_2 (Primary Slot Run Stoppers, e.g., Cooper Dejean). Safeties were split between S_2 (Dynamic Ball-Hawking Deep Safeties) and S_3 (Run-Stopping Slot/Box Safeties).



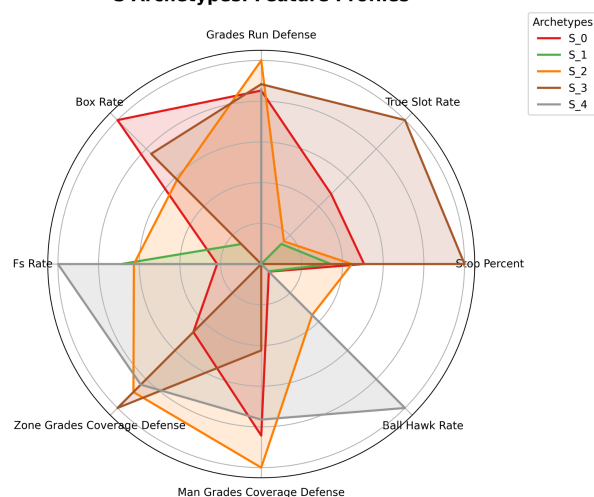
LB Archetypes: Feature Profiles



CB Archetypes: Feature Profiles



S Archetypes: Feature Profiles



3.2 Tactical Valuation: Linear Mixed-Effects Modeling

A primary challenge in evaluating defensive performance is the inherent bias of schedule strength; a defensive lineup may generate a highly favorable Expected Points Added (EPA) simply because it faced a statistically inferior opposing offense. To correct for this bias and isolate the pure tactical value of specific personnel groupings, this study employed a Linear Mixed-Effects Model using the Python statsmodels library.

The dependent variable was play-level EPA. To ensure sufficient sample size and model convergence, the dataset was filtered to include only 11-man archetype combinations deployed for at least 50 snaps over the five-year sample. The 11-man archetype lineup (e.g., 1 DI_1, 2 ED_2, 2 CB_1, etc.) was treated as a categorical fixed effect. Furthermore, contextual game-state variables – down, distance (yards to go), and field position (yardline) – were included as fixed covariates to control for the defense's situational advantage.

Crucially, the opposing offensive team (posteam) was introduced as a random intercept. This random effect computationally "forgives" a defense for yards yielded to elite offenses, while penalizing them for struggling against lower-tier units. The resulting coefficients from this model represent the **Opponent-Adjusted EPA** of each specific 11-man package, providing a rigorous, isolated metric of tactical efficiency.

Opponent-Adjusted EPA Formula:

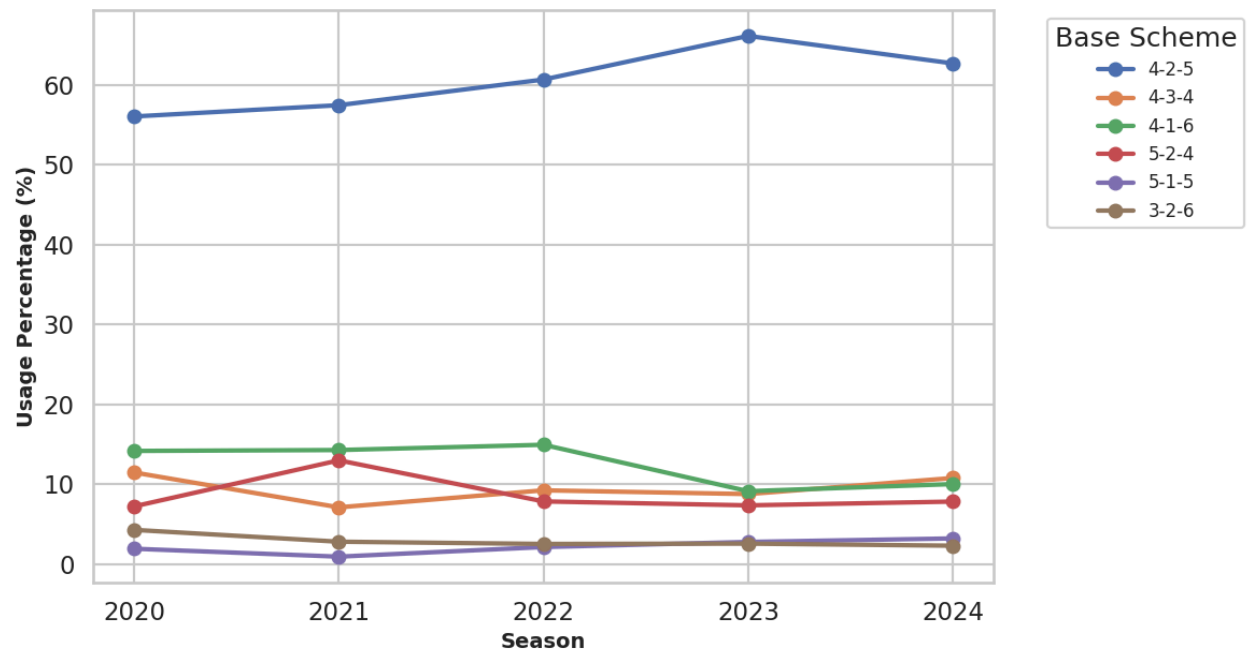
$$\text{EPA} = \beta_0 + \beta_1(\text{Lineup}) + \beta_2(\text{Down}) + \beta_3(\text{Distance}) + \beta_4(\text{Field_Position}) + \text{Random_Intercept}(\text{Opponent}) + \varepsilon$$

4. Results

4.1 The Death of the Base Defense

The data overwhelmingly demonstrates that traditional base defenses are mathematically sub-optimal in the modern NFL. Longitudinal deployment tracking over 2020-2024 reveals a steady, sustained decline in the use of 4-3 and 3-4 defensive structures. Concurrently, the 4-2-5 Nickel defense generated consistently more favorable (negative) adjusted EPA values, driving its adoption as the unquestioned standard base defense across the league.

Evolution of NFL Defensive Base Schemes (2020 - 2024)



4.2 The Optimal 11-Man Blueprint

After adjusting for the strength of the opposing offense, the Linear Mixed-Effects Model revealed significant variances in the tactical efficiency of 4-2-5 Nickel packages. Of the qualifying lineups, approximately 7% had statistically significant p-values, indicating that their success was driven by personnel synergy rather than random variation.

	Archetype_Lineup	Base_Scheme	Plays	Avg_EPA	Std_Error	P_Value
0	CB_1_CB_4_CB_4_DI_0_DI_1_ED_0_ED_1_LB_1_LB_1_S...	4-2-5	50	-0.460990	0.214639	0.031734
1	CB_4_CB_4_DI_2_DI_2_DI_3_ED_1_LB_1_LB_1_LB_2_S...	4-3-4	56	-0.368084	0.202663	0.069334
2	CB_2_CB_4_CB_4_DI_0_DI_0_ED_0_ED_1_LB_1_LB_1_S...	4-2-5	66	-0.366356	0.189044	0.052630
3	CB_2_CB_2_CB_4_DI_0_ED_0_ED_0_ED_1_LB_3_S_1_S...	4-1-6	94	-0.320658	0.166165	0.053637
4	CB_2_CB_4_CB_4_DI_0_DI_0_ED_0_ED_1_LB_1_LB_1_S...	4-2-5	66	-0.231040	0.189574	0.222947

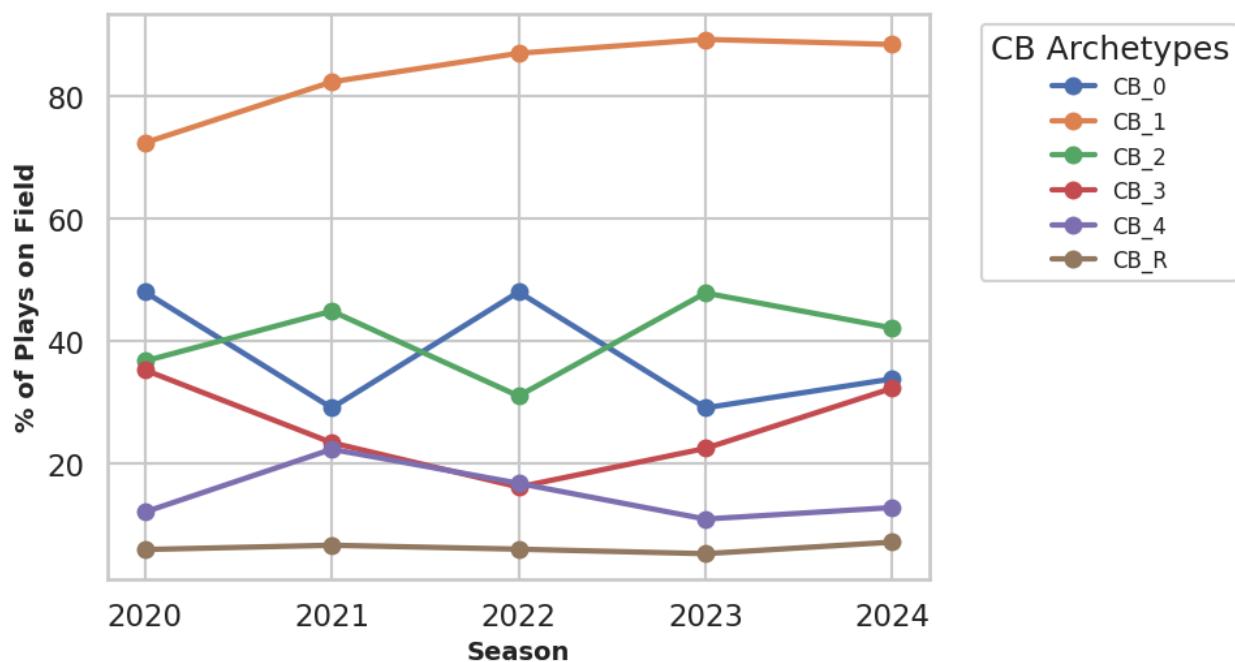
The most dominant personnel grouping identified by the model generated an Opponent-Adjusted EPA of **-0.46 per play**. This elite configuration pairs an interior pass-rush specialist (DI_1) and high-level playmakers in the defensive backfield—specifically, two man-coverage ball-hawks (CB_4) and two dynamic ball-Hawking deep safeties (S_2). This confirms that modern defensive dominance relies on highly specific "Inside-Out" pairings. The data suggests that optimal defensive success is achieved by prioritizing interior disruption and deep-ball turnover generation over traditional, heavy run-stopping roles.

4.3 The Positional Pivot: Defensive Backs

The year-over-year archetype trends unearth a clear league-wide priority shift toward deep-coverage over run support:

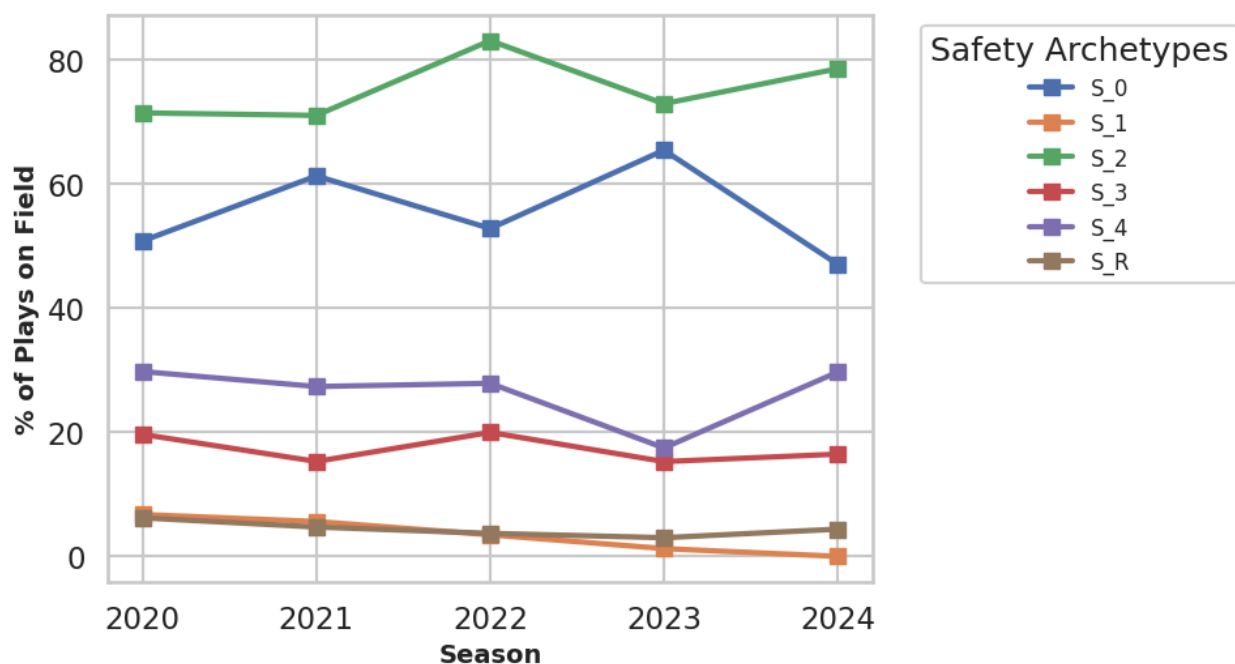
- **Cornerbacks:** It was initially hypothesized that defenses would increase the deployment of specialized run-stoppers (CB_2) to counter the spread. However, the data revealed a different evolutionary path. CB_2 usage stagnated at ~40%. Conversely, the CB_1 (Traditional Outside Zone Corner) climbed to an overwhelming ~90% usage rate by 2024.

Evolution of Cornerback Archetypes (2020 - 2024)



- Safeties:** Similar trends were observed at the safety position. Box safety usage (S_3) remained entirely flat at roughly 15-20%. In stark contrast, the Dynamic Ball-Hawking Deep Safety (S_2) saw substantial growth, reaching nearly 80% usage by the final tracked season.

Evolution of Safety Archetypes (2020 - 2024)



5. Discussion and Conclusion

By applying Gaussian Mixture Models to five years of NFL tracking and grading data, this study successfully quantified the tactical value of specific defensive schemes and player archetypes. Furthermore, by implementing a Linear Mixed-Effects Model, this study isolated the Opponent-Adjusted EPA for specific defensive packages, filtering out noise from schedule strength.

The findings mathematically prove that the traditional 4-3 base defense has been practically replaced by the 4-2-5 Nickel. Furthermore, the granular analysis of archetype trends reveals a league-wide pivot toward deep-coverage and boundary zone specialists over traditional box defenders. As offenses continue to optimize passing efficiency, defenses have statistically concluded that preventing explosive passes is paramount, leading to a profound shift in the types of athletes prioritized on the field.

The findings of this tactical analysis naturally inform the next frontier of this research: cost-efficiency. Now that the highest-performing defensive archetypes have been successfully identified and validated via EPA, future analysis can examine salary allocations for these specific sub-packages. By merging these tactical insights with salary cap data, subsequent research can answer whether NFL front offices are efficiently allocating their financial resources toward the modern defensive archetypes that actually drive on-field success.